

Longitudinal Analysis of Sepsis-3 Onset in the ICU: Toward Real-Time Prediction (MIMIC-IV)

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1 Introduction

A companion cross-sectional report analysed each ICU stay as a **single row** of first-24h summaries and asked “*was this stay septic?*” That framing cannot address the project’s actual objective: a **real-time early-warning system** that, at each hour and using only the data observed so far, estimates the risk that a patient will develop sepsis. This report rebuilds the problem on a **longitudinal** footing.

From the raw tables a pipeline (scripts 01 to 08) resamples every stay onto an **hourly grid**: a panel of **3,095,002** stay-hours across **65,241** stays (mean 47 h/stay). SOFA is scored every hour, and the hour of **sepsis onset** is located as the first hour, inside the suspected-infection window, at which SOFA reaches the organ-dysfunction threshold. This gives what the static matrix never had: a **time axis** and an **onset time**.

Two facts shape everything that follows:

1. **32.0%** of stays meet Sepsis-3 within the 72-hour horizon, but of these **34%** are already septic at ICU admission (onset hour 0): *prevalent*, not predictable. The median onset is hour **2**.
2. The clinically useful question is therefore about **incident** sepsis: among patients *not yet* septic, who will become so? We answer it with a **landmark design** at hour **6**: conditioning on the first six hours and predicting onset thereafter, which is exactly how a deployed model would be used.

The methods are the longitudinal counterparts of the cross-sectional report: **Kaplan-Meier** and **Cox** (including a **time-varying Cox**) for time-to-onset, **Fine-Gray competing risks** (death competes with onset), **autocorrelation and trajectory** analysis, a **within-patient hidden Markov model** (the HMM applied to real sequences, not stay order), and **time-dependent discrimination** benchmarked against the streaming clinical scores a new model must beat.

2 The Hourly Panel and Onset Labelling

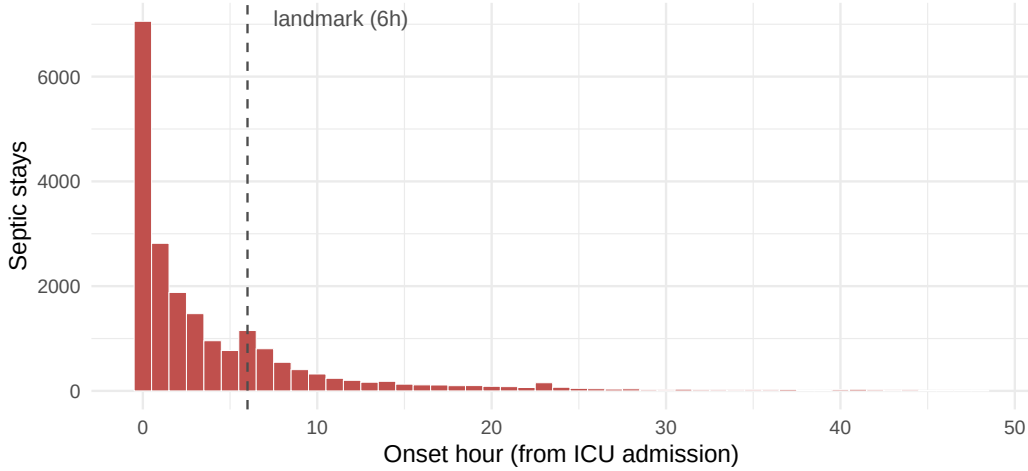


Figure 1: Distribution of sepsis-onset hour among septic stays. A large spike at hour 0 marks sepsis already present at admission (prevalent); the right tail is incident sepsis developing in the ICU: the target of early prediction.

Figure 1 shows the split between prevalent and incident sepsis. Conditioning at the hour-6 landmark leaves **48,829** stays still at risk, of whom **4,750** (9.7%) later develop sepsis and **3,330** (6.8%) within the next 12 hours. This is a realistically **imbalanced** early-prediction problem, unlike the near-balanced stay-level label.

3 Time-to-Onset: Kaplan-Meier

Time-to-event methods treat onset as an event that may or may not have occurred by the end of observation (censoring), which a simple classifier ignores. The survivor function $S(t)$ gives the probability of remaining sepsis-free to hour t ; $1 - S(t)$ is the cumulative incidence.

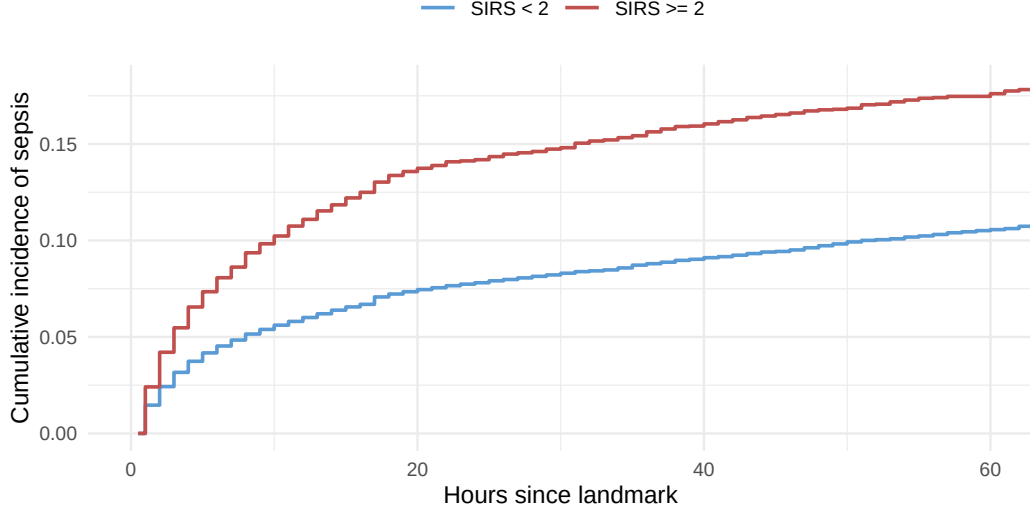


Figure 2: Cumulative incidence of sepsis from the hour-6 landmark, stratified by SIRS at the landmark. Patients meeting ≥ 2 SIRS criteria reach sepsis substantially faster.

Figure 2 shows the cumulative incidence separating early by SIRS status: by 24 h after the landmark it reaches roughly **14.2%** in the $\text{SIRS} \geq 2$ group versus **7.8%** otherwise. A log-rank test confirms the difference.

The log-rank statistic is **360** ($p < 1\text{e-}16$).

4 Cox Regression and the Failure of Proportional Hazards

A Cox model estimates how baseline (hour-6) covariates scale the hazard of onset. Hazard ratios (HR) above 1 accelerate onset.

Table 1: Cox model for time to incident sepsis (hour-6 landmark).

Term	HR	95% CI	p
age	1.009	[1.007, 1.011]	$< 1\text{e-}16$
hr	1.011	[1.009, 1.013]	$< 1\text{e-}16$
resp_rate	1.021	[1.019, 1.023]	$< 1\text{e-}16$
map	0.988	[0.986, 0.990]	$< 1\text{e-}16$
sofa_total	0.762	[0.748, 0.777]	$< 1\text{e-}16$
sirs	1.150	[1.104, 1.198]	$2.5\text{e-}11$

Table 1 shows **SIRS** and the **vital signs** carry the hazard, while a higher **single-timepoint SOFA** is associated with a *lower* hazard of *incident* sepsis (HR 0.76): patients with organ dysfunction but no evolving infection at hour 6 tend not to cross the Sepsis-3 threshold later. The global test of the **proportional-hazards assumption fails** ($p < 1\text{e-}16$), indicating that the covariate effects **change over time**, a violation that a baseline Cox cannot express and that motivates a time-varying model.

4.1 Time-varying Cox

The hourly panel is cast as a counting process with start-stop intervals, allowing covariates to **update every hour** so the model uses each patient’s *current* physiology rather than a fixed baseline.

Table 2: Time-varying Cox: hazard of onset from hour-updated physiology (sampled stays).

Term	HR	95% CI	p
sofa_total	0.801	[0.729, 0.880]	3.6e-06
hr	1.019	[1.010, 1.028]	1.4e-05
map	0.993	[0.982, 1.004]	0.21
lactate	0.991	[0.811, 1.211]	0.93

Table 2 updates every covariate hour by hour. The **current heart rate** is positively associated with imminent onset (HR 1.019 per bpm), whereas the **single SOFA value stays inversely associated** (HR 0.80): organ dysfunction without an evolving infection does not progress to Sepsis-3, whether read at admission or hour-by-hour. Allowing covariates to vary in time rescues the violated proportional-hazards assumption, and the predictive signal lives in **physiological dynamics (notably heart rate)** rather than in a severity snapshot, the same conclusion the discrimination analysis reaches. (With 161 onset events in the sampled counting process, these hourly hazards are directional rather than precise.)

5 Competing Risks

Death and discharge **compete** with sepsis onset: a patient who dies sepsis-free can never develop sepsis, so treating those as ordinary censoring overstates incidence. The **Fine-Gray** model targets the cumulative incidence of onset in the presence of the competing event.

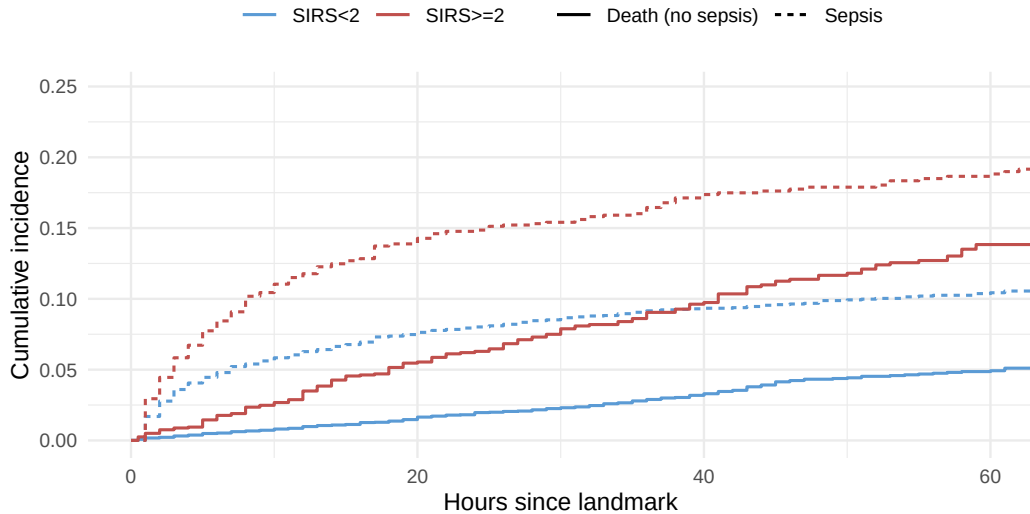


Figure 3: Cumulative incidence functions from the landmark, accounting for death without sepsis as a competing risk, by SIRS status.

Figure 3 shows onset incidence (solid) rising faster in the $\text{SIRS} \geq 2$ group, while the competing risk of death without sepsis (dashed) is modest over the horizon. The Fine-Gray subdistribution hazard ratio

for SIRS is **1.49**, confirming SIRS raises the *cumulative* likelihood of onset even after allowing for the competing event.

6 Temporal Structure: Trajectories and Autocorrelation

The cross-sectional report’s HMM disclaimer noted that stays “are not a true time series.” Within a patient, however, the hourly measurements emphatically *are*.

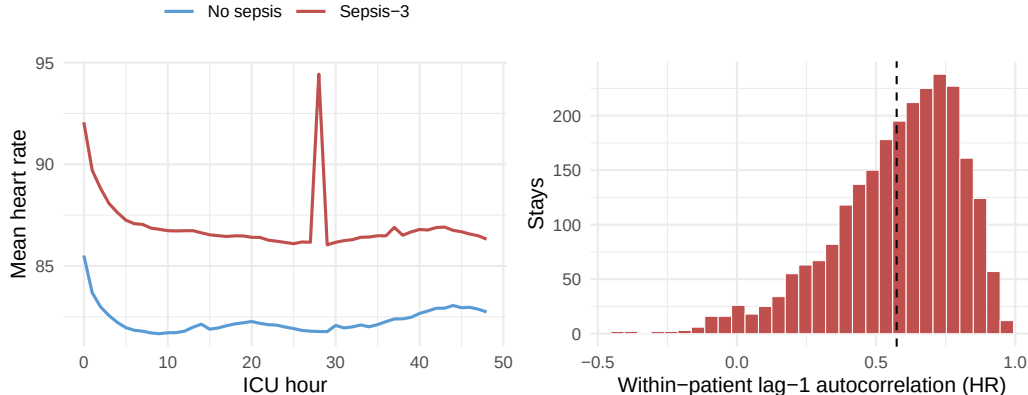


Figure 4: Left: mean hourly heart rate and SOFA by sepsis status (aligned to admission): septic patients run higher and diverge. Right: distribution of within-patient lag-1 autocorrelation of heart rate; values well above zero show strong hour-to-hour dependence.

Figure 4 shows septic patients running at a higher mean heart rate throughout, and a mean within-patient lag-1 autocorrelation of heart rate of **0.57**: strong serial dependence. This dependence is the statistical foundation the static analysis discarded and that sequence models are built to exploit.

7 Within-Patient Hidden Markov Model

The HMM is now applied to each patient’s **hourly heart-rate sequence**, where a latent state genuinely persists and transitions over time. A three-state Gaussian HMM is fitted jointly across a sample of stays (`ntimes` = each stay’s length).

Table 3: Three-state HR HMM: emission means, diagonal (persistence) of the transition matrix, and the fraction of hours in each state that are labelled septic.

State	HR mean	Self-persistence	Septic-hour rate
1 (low)	62.6	0.97	33.0%
2 (mid)	80.8	0.95	37.0%
3 (high)	102.6	0.96	45.3%

Table 3 shows the three states order into low, medium and high heart-rate regimes with **strong self-persistence** (diagonal transition probabilities around 0.96): genuine temporal stickiness, unlike the artefactual diagonal the cross-sectional HMM produced on stay *order*. The high-HR state also carries the highest proportion of septic hours, so the latent state tracks physiological deterioration. This is the HMM doing real work: modelling how a patient moves between hidden physiologic regimes over time.

8 Time-Dependent Discrimination and Clinical Baselines

This section establishes the **performance bar** a real-time model must clear: how well do the standard streaming scores predict incident sepsis, and does using the *trajectory* help?

Table 4: Discrimination for incident sepsis within 12h of the landmark (cross-validated for the multivariable models).

Score	AUC (within 12h)
Single clinical scores	
SOFA (hour 6)	0.359
qSOFA (hour 6)	0.492
SIRS (hour 6)	0.586
Multivariable	
Multivariable (static, 6h)	0.749
Multivariable + 6h trajectory slopes	0.755

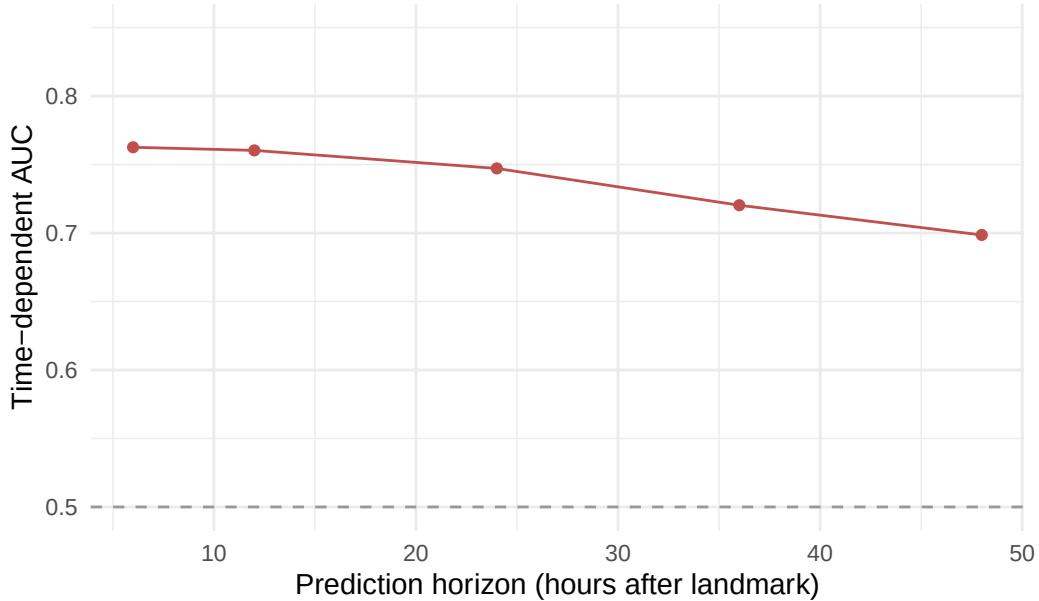


Figure 5: Time-dependent AUC for predicting sepsis onset over increasing horizons, using the cross-validated multivariable risk score.

Table 4 is the central finding. **No single clinical score is useful** for incident sepsis: SOFA is even *inverse* (AUC 0.359), because organ dysfunction without evolving infection does not progress to Sepsis-3; SIRS is the best single score (AUC 0.586). A **multivariable** combination of the same hourly measurements reaches AUC **0.749**, and adding **six-hour trajectory slopes** raises it further to **0.755** (DeLong p 3.5e-07). Figure 5 shows this discrimination holds across horizons. The temporal features add real, statistically significant signal, providing direct evidence that the trajectory carries information a snapshot does not.

9 Conclusion and Path to a Real-Time Model

Reframing the problem longitudinally changed both the methods and the findings.

- **Prevalent vs. incident.** A third of Sepsis-3 is present at admission and not predictable; the real task is incident sepsis, isolated by the hour-6 **landmark** (9.7% of at-risk stays).
- **The signal is in the dynamics, not the snapshot.** A baseline Cox violated proportional hazards; a **time-varying Cox** located the hazard in *current heart rate* rather than a SOFA severity score (which remains inversely related to incident onset). **Autocorrelation** (0.57) and a persistent-state **HMM** confirmed genuine within-patient temporal structure that the cross-sectional analysis had to discard.
- **A performance bar.** Single streaming scores are weak or inverse; a multivariable hourly model reaches AUC **0.749** and trajectory features push it to **0.755**. Any real-time model should be judged against these, with time-dependent AUC and competing-risk-aware, leakage-safe, patient-grouped evaluation.

Toward the real-time model. The labelled hourly panel, onset times, landmark design, competing-risk framing, and streaming baselines assembled here are the direct inputs to the real-time statistical model built next (scripts 10-19): a dynamic **landmark supermodel** fit entirely with penalised logistic regression, GEE, and time-varying Cox (no gradient boosting or neural sequence models), emitting an hourly sepsis risk by stacking predictions across a grid of landmark times. The statistics in this report specify the target, the evaluation, and the minimum performance that model must exceed.

Limitations. Onset depends on clinician behaviour (when cultures and antibiotics are ordered), so incident “onset” partly reflects *recognition* timing; the SOFA baseline is assumed zero; qSOFA uses MAP as a proxy for systolic pressure; and the time-varying Cox and HMM use a sampled subset for tractability. These are revisited as the modelling matures.

Acknowledgment

This longitudinal analysis extends the cross-sectional study to the project’s real objective: real-time prediction, using the **PAIR framework** for responsible AI use: the problem framing, survival and time-series methods, and their honest (including negative) interpretation were critically reviewed against primary sources.

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